PID-RS TECHNICAL REPORT SERIES

Ecosystem Compatibility Audit

pid-rs, Prisoma, Galadriel, Haldir, and Crebain

pid-rs project

25 July 2026

Abstract

This source-read audit compares the recorded source states of Prisoma, Galadriel, Haldir, Crebain, and pid-rs. It separates implemented behavior, proposed behavior, stale documentation, unsupported assumptions, and missing evidence. The reviewed consumer sources contain no qualification evidence for the pid-rs source base reviewed here. The strongest workable route is an offline categorical or independently frozen quantized analysis with explicit target, row, transform, support, selection, and uncertainty contracts. Continuous KSG and shared-exclusions PID remain restricted research routes. PID evidence may never grant or widen mission or plant authority. Exact XOR, coherent-majority, and target-rotation constructions show why pairwise agreement and changing-target atom reports cannot carry stronger interpretations.

Contents

1 Claim boundary and source control

This report is a compatibility review. It is not scientific qualification, deployment approval, formal refinement, or an authority claim. A local checkout does not prove that the same bytes were reviewed, built, or deployed elsewhere.

The machine-readable file `ecosystem-capabilities.json` remains the authority for its historical capability-and-gap projection. This paper has a different purpose. It reads the present local consumer sources and records concrete integration findings. It is not a generated contract.

No repository was missing. Three source bases were not clean, including two consumer checkouts. Table 1 records the exact state observed.

Table 1: Source bases read during the audit.

Repository	Revision and branch	State	Evidence consequence
pid-rs	`67e8b92d42d23a051b839bc81ba05c26c08377fe`, main	30 pre-existing entries	The live contracts were read. This is not an immutable release snapshot.
Prisoma	`879a1909a986bda381f86979b5c02e3cec89d68c`, main	clean	Findings bind this local commit.
Galadriel	`80506dd2ce52b33c3334c7d1760a8155c7631241`, main	64 entries	Relevant galadriel-pid changes were documentation wording. The full checkout is not immutable evidence.
Haldir	`bb6c0a7b27bbc57fe9935f80e22d06ca3b60e8ba`, `wip/current-file-review-ledger`	12 entries	The two cited evidence files were unchanged. This is not a main-branch release claim.
Crebain	`f1f4c050a1aa6f703b1b25699c232f4305bfb001`, main	clean	Findings bind this local commit and the current-head refresh below.

A release qualification must repeat this review on clean, pinned commits and retain hashes of every source, configuration, input, transform, and output.

Current-head Crebain refresh. Crebain was rechecked at the clean commit above against the previously audited base `448c9c43c15b0081c879037aa62b6a9f1bc4a341`. The three-commit delta changes 12 dependency, security, baseline, and CI files (171 insertions and 93 deletions), and `git diff --check` passed. Manual review of the complete delta found no change to PID/SxPID behavior, the Galadriel scientific data contract, sensor-fusion algorithms, or the authority boundary. In a detached clean worktree, a frozen-lockfile install followed by `bun run validate:all` exited zero. This included 551 frontend tests; 383 passed and one ignored default Rust library tests plus 22 benchmark tests; and 485 passed and one ignored NCP-feature Rust library tests plus 22 benchmark tests. The original checkout's ignored `.env` correctly caused the Phase 0 baseline gate to reject ambient state. It was neither read nor moved; the detached worktree was used instead. These are source and component-validation results, not a release qualification, live receiver or deployment test, statistical calibration, or end-to-end PID evidence.

2 Review method

The audit used a claim-to-code trace rather than a keyword count:

1. Read each available repository instruction file and recheck source interpretations against it.
2. Record branch, commit, dirty state, dependency pins, and submodule pins.
3. Trace data from producer or adapter through preprocessing and estimator selection to terminal status and any authority boundary.
4. Prefer executable source and manifests over summaries; mark a document stale when it contradicts current code.
5. Challenge operational rules with exact negative constructions. Direct enumeration of four equiprobable binary states verifies the XOR and target-rotation identities. The coherent-majority result uses the closed-form Gaussian MI formula.
6. Search manifests and Rust sources before reporting that no direct dependency was found.
7. Do not treat component tests, declared assumptions, or producer publication as end-to-end scientific evidence.

Apart from the current-head Crebain source/component validation above, the audit did not run consumer release qualification, a live Neuro-Cybernetic Protocol (NCP)/Galadriel deployment, or a statistical calibration study. Companion PDF build and render checks are a separate presentation-integrity evidence layer; they do not fill those scientific or operational gaps.

3 Executive determination

No qualification evidence against the pid-rs source base reviewed here was found in the reviewed consumer source bases.

Table 2: Current ecosystem determination.

Consumer	Implemented behavior	Determination	Principal blocker
Prisoma	Direct path dependency through a pinned submodule; continuous and discrete I_{min} screens	Partly integrated; not scientifically qualified	SAFE omits the harness support map. Same-row PLS and quantization are descriptive. No confirmatory holdout is registered.
Galadriel	Direct immutable pid-core pin; pairwise-KSG clique verdict; report-only continuous I^{sx} atoms	Advisory research integration only	The verdict is not a PID verdict. Pure synergy, coherent-majority inversion, changing targets, and sequential calibration remain unresolved.
Haldir	No direct pid-rs dependency found; design/review document only	Not integrated; permanent no-authority boundary	The design note contains stale API and uncertainty claims. No executable adapter exists.
Crebain	Optional Galadriel-compatible evidence producer; no direct pid-rs dependency found	Producer plumbing only	No live receiver, window assembler, or PID adapter exists. The envelope lacks the scientific and estimator contract.

Table 3 makes the lifecycle of each claim explicit.

Table 3: Implementation, plan, stale statement, and unsupported interpretation.

Consumer	Current implementation	Planned or missing interface	Stale statement	Unsupported interpretation
Prisoma	Pinned continuous and discrete I_{min} harness	H3 scorer, holdout, frozen transform, updated schema	SAFE says the schema mirrors Rust exactly	Same-row fitting or asserted support as confirmation
Galadriel	Pairwise-MI clique plus report-only atoms	Fleet and repeated-window calibration; fixed-target report	Broad PID engine/verdict naming	Pure synergy, trustworthy majority, causal attribution
Haldir	No executable adapter found	Advisory Mirror receiver and evidence contract	No multiplicity correction; jitter repair; raw "CI"	PID authority, alarm-grade latency, calibrated posterior
Crebain	Optional bounded producer	Receiver, window receipt, scientific adapter	Historical revision records as current proof	Publication as end-to-end PID, calibration, or authority

The usable route is narrow. Low-cardinality categorical or independently frozen quantized SxPID can support offline held-out analysis when target, row law, transform, support, uncertainty, and selection are explicit. Continuous KSG and ISX remain restricted research routes. Neither route supplies mission authority.

3.1 Terminology

The phrase "colored PID" does not name a PID measure. Dependency coloring qualifies a finite-sample concentration theorem for categorical SxPID under a specified row-dependence contract. It constrains dependence among sample rows. It is not a new PID functional, estimator, or term attributed to Makkeh, Gutknecht, or Wibral. Consumer documentation should use *dependency-colored concentration for categorical SxPID*.

The executive rows are not single-lens conclusions. Each determination is justified by the seven-lens assessment below. A route fails qualification if any required lens remains unsupported.

4 Mandatory seven-lens assessment

4.1 Prisoma

Provenance
: Prisoma pins pid-rs at `796c11e70f009634b853dc4ada6f565563d82f51`, not the pid-rs source base reviewed here. The consumer checkout was clean, but compatibility with current pid-rs is not established.

SxPID estimand compatibility
: Current discrete routes are labelled I_{min}, whose redundancy definition differs from shared

exclusions. Continuous ISX is a different estimator and regime. No measure-specific I_{min} atom value, theorem, sign property, or benchmark transfers to SxPID without an explicit mapping theorem.

Mathematical assumptions

A usable route needs a fixed target, disjoint transform fitting, a justified finite alphabet or declared continuous support, a valid row-dependence model, and a predeclared selection family. The current support value is asserted by the caller, not proved by data.

Implementation and API

SAFE omits the Rust harness's top-level `support` map. Same-row PLS and quantizer fitting remain descriptive. The holdout registry contains no confirmatory holdout.

Numerical and statistical validity

Missing support causes typed continuous abstention. KSG/ISX geometry gates do not establish consistency in the application law. Same-row fitting has no out-of-sample calibration, and raw resampling ranges are not generic PID confidence intervals.

Adversarial and operational misuse

Independent high-dimensional noise can be target-fitted to interpolate evaluation labels, yielding maximal training-row association and zero MI under the test law. An asserted support label can also turn a model assumption into an apparently measured fact.

Authority

Prisoma results may support a preregistered offline scientific analysis only. They cannot authorize deployment, select a model on confirmatory rows, or convert abstention into evidence of no effect.

4.2 Galadriel

Provenance

Galadriel pins pid-core at `1cd2424f7967e1752dcc8e53859e8fdad3566f51`, not the pid-rs source base reviewed here. The checkout was dirty. Relevant galadriel-pid differences were documentation wording, but the tree is not immutable qualification evidence.

SxPID estimand compatibility

The operational verdict is a pairwise-MI clique rule. Continuous SxPID atoms are report-only. Atom rows change the target, so they are not coordinates of one fixed decomposition. Pairwise MI, I_{min}, categorical SxPID, and continuous ISX are not interchangeable.

Mathematical assumptions

The route needs regular full-dimensional continuous laws, finite MI, valid row sampling, common scalar projections, a fixed scientific target for atom comparison, and justified thresholds. Added pseudo-noise proves none of these assumptions.

Implementation and API

PID is optional and off by default. The engine computes gated pairwise MI, selects a unique strict-majority clique, and separately reports atoms. Legacy `Jitter` changes the analysed law and lacks a complete application receipt.

Numerical and statistical validity

Geometry and shell checks are one-sided diagnostics, not calibration. Delete-block stability is not a generic confidence interval. No fleet-level, repeated-window, post-selection, or false-alarm guarantee covers the clique threshold or attribution.

Adversarial and operational misuse

XOR is invisible to the pairwise graph; a coherent faulty majority can isolate the honest channel; and rotating targets invalidate cross-row atom comparison. Common-mode faults require no adversary and produce the same majority failure.

Authority

Galadriel's rules correctly keep PID advisory: nominal cannot create permission and insufficient evidence cannot become nominal. No MI edge, clique, atom, or score may grant or widen authority.

4.3 Haldir

Provenance
: No direct pid-rs or pid-core dependency was found. The reviewed PID material is a design and adversarial note on a dirty non-main branch; the cited note and frozen authority file were unchanged.

SxPID estimand compatibility
: No executable SxPID estimator or adapter exists. The note distinguishes $I_{\min}$ from I^{sx}, and that distinction must remain. Neither definition validates the other without a mapping theorem for the declared distribution class and atom semantics.

Mathematical assumptions
: Sequential monitoring, adaptive windows, drift, dependence, missingness, transform selection, and post-selection need explicit theorems or validation. A fixed-window estimator theorem does not establish alarm or authorization performance.

Implementation and API
: The proposed Mirror path is not implemented. The note has stale multiplicity claims, recommends generic jitter repair, and labels uncalibrated resampling ranges as confidence intervals.

Numerical and statistical validity
: Current pid-rs BH/BY functions require valid p-values and their stated dependence conditions. Block sensitivity ranges are not generic KSG/PID intervals. Sample-size/latency conflict and slow-drift blind spots remain unresolved.

Adversarial and operational misuse
: A coherent wrong majority, slow drift, missing evidence coerced to zero, or a score-to-command bridge can defeat the intended interpretation. Statistic verification does not establish reference truth or causal attribution.

Authority
: Any PID-aware authorization must imply the original authorization. PID may restrict an already allowed action under a predeclared rule; it may never create an allow, override a failed conjunct, or route a command.

4.4 Crebain

Provenance
: The clean checkout has no direct pid-rs dependency. Historical revision records and a successful local producer put do not prove receiver receipt, accepted schema, estimator identity, or deployed compatibility.

SxPID estimand compatibility
: `PidObservation` is an envelope name, not an SxPID definition. It does not bind a target, ordered sources, measure, lattice, units, or atom semantics. NIS and consistency projections cannot be transplanted into PID quantities.

Mathematical assumptions
: A valid adapter needs a fixed frame/context/horizon, row unit, receiver-window construction, missingness and drift accounting, support declaration, transform split, and source/target semantics. These fields are absent from the current envelope.

Implementation and API
: The optional producer is off by default and needs exact runtime opt-in. No live receiver, window assembler, registry transform executor, or pid-rs adapter was found. Projection is

emitted only for an already matching frame and empty transform chain.

Numerical and statistical validity
: Producer bounds and NIS fields do not calibrate any PID estimator. Publication does not establish delivery or acceptance. No estimator configuration, typed abstention, interval status, multiplicity family, or run-log replay receipt accompanies the observation.

Adversarial and operational misuse
: Loss, reorder, duplicates, restart, saturation, target rotation, coherent-majority data, and registry disagreement can change a downstream window or estimand while the producer remains locally successful.

Authority
: Crebain's trust-domain table correctly gives Galadriel advisory observation and no command capability. Producer, receiver, observer, Haldir, command, and plant credentials must remain separate.

5 Minimal mathematical primer

For discrete random variables, mutual information is

$$I(X;T) = \sum_{x,t} p(x,t) \log \frac{p(x,t)}{p(x)p(t)}. \tag{1}$$

It is zero exactly when X and T are independent. It measures statistical dependence, not causality, truth, intent, or fault.

A two-source PID resolves the total information as

$$I((S_1, S_2);T) = \mathrm{Red} + \mathrm{Unq}_1 + \mathrm{Unq}_2 + \mathrm{Syn}\,. \tag{2}$$

The target T is part of the definition. Changing T changes the decomposition. Shared-exclusions PID also separates informative and misinformative pointwise components; signed net atoms can be negative. The $I_{\min}$ measure of Williams and Beer uses a different redundancy definition and must not be reported as shared-exclusions PID.

Categorical routes operate on finite empirical count tables. KSG and continuous ISX use local-neighbourhood geometry and require stronger support, sampling, dimension, and tie assumptions. Correct formulas and clean component tests do not establish those application premises.

5.1 No-transplant rule

Let claim C_M be proved or validated for a measure or estimator M. It may be used for another measure or estimator N only if an explicit mapping theorem states the distribution class, target and source semantics, lattice coordinates, normalization and units, numerical representation, and the property preserved by the map. Shared atom names, a common reconstruction identity, similar benchmark values, or use of the same MI terms are not such a theorem. This review found no mapping theorem in the checked source that licenses any transfer below.

Consequently:

- $I_{\min}$ results do not validate SxPID atoms;
- categorical SxPID theorems do not validate continuous ISX estimators or claims about an unquantized law;
- KSG MI checks do not validate continuous PID atom bias, variance, or coverage;
- two-source results do not validate three-source or mixed-dimensional lattices; and

- a pairwise-MI consensus rule does not become a PID decision rule because PID atoms appear in the same report.

When no mapping theorem exists, the adapter must report separate estimands and separate evidence.

6 Minimum cross-consumer contract

An adapter must retain the fields below. A label without evidence is not a certificate.

6.1 Identity

- Consumer repository, commit, dirty state, build features, and executable identity.
- Exact pid-rs revision, crate version, API revision, method-catalog entry, and catalog digest.
- Reported units: pid-rs information quantities use natural logarithms and nats.
- Exact estimator family. I_{min}, categorical I^{sx}, quantized I^{sx}, and continuous I^{sx} are different routes. One is not a silent fallback for another.

6.2 Scientific roles

- One target and one ordered, labelled source tuple.
- Physical meaning, units, common frame, and horizon for every variable.
- Row unit: frame, episode, track window, trial, subject, or another declared unit.
- Population or finite-population estimand. A window statistic is not automatically a fleet or trajectory estimand.

6.3 Sampling and support

- IID, stationary, exchangeable-block, dependency-color, or unsupported row status.
- For a dependency coloring, a deterministic color map and a justification of *mutual* independence of complete rows within a color. Pairwise uncorrelatedness is not enough.
- Drift and selection declarations. Concentration about an average row law does not identify one fixed target law without a drift bound.
- A categorical alphabet identity or a continuous/mixed support declaration. Unique observed rows cannot prove a regular full-dimensional population law.
- A population support floor only when a theorem requires it. An observed minimum cell mass cannot replace an unknown population floor.

6.4 Preprocessing

- Training and evaluation row identities and disjointness at the episode or subject level.
- Frozen transform parameters or bytes, source order, training digest, and application digest.
- Target-adaptive selection and fitting outside the evaluation rows.
- Quantizer edges as part of the estimand. Same-row bins define a descriptive, data-adaptive quantity.
- Added noise only under a declared observation model or labelled sensitivity study. Noise changes the estimated law; it is not a generic tie repair.

6.5 Estimator, status, and replay

- Metric, neighbour count, gauge, seeds, resources, and signed negative handling.
- Distinct `produced`, `warning`, `abstained`, and `failed` outcomes. Invalid output must not become zero.
- Signed informative, misinformative, and net atoms where available. A negative net atom is not a causal, fault, intent, or responsibility finding.
- Exact interval status: numerical enclosure, calibrated confidence interval, raw resampling percentile, or sensitivity envelope.
- A predeclared testing family and valid p-values before BH or BY.
- Input, transform, configuration, output, run-log, and software-identity digests, including terminal failures.
- An explicit advisory-only/no-authority marker.

7 Prisoma

7.1 Implemented route

Prisoma consumes the pid-rs submodule at `796c11e70f009634b853dc4ada6f565563d82f51`. The harness records this revision in `crates/pid-sim/src/offline_harness.rs`, lines 3937–3938. This pin predates the pid-rs source reviewed here. Current-main compatibility is not established.

The harness implements continuous KSG/ISX screens under a caller support assertion, discrete and discrete-PLS $I_{\min}$ screens, fitted equal-width quantization, and typed abstention. Its own documentation correctly says that `assume_regular_full_dimensional` is an assertion, not proof (lines 4276–4286).

7.2 Confirmed SAFE schema defect

The Python SAFE adapter states that it mirrors the Rust dataset exactly. It does not. The Rust `OfflineVldaDataset` has a top-level support map. A missing axis fails closed as `support_contract_unspecified` (`crates/pid-sim/src/offline_harness.rs`, lines 369–378, 3964, and 3998). The Python `VldaDataset` has only samples and four optional text fields. Its serializer emits no support map (`experiments/safe_adapter/contract.py`, lines 147–211). The converter and verifier also omit it.

Thus a JSON object can pass the Python verifier and make every continuous Rust tuple abstain. This is a reproducible cross-language schema incompatibility. The repair is concrete:

1. Add closed support declarations for `v`, `l`, `d`, and `a`.
2. Reject absent, unknown, or contradictory declarations in Python.
3. Preserve capture integrity and publication receipts when required.
4. Retain an old-schema negative fixture that must end in `support_contract_unspecified`.
5. Remove the word “exactly” until a cross-language golden test proves it.

7.3 Same-row fitting is descriptive

Discrete-PLS fits each source toward the target on the rows supplied to the screen and transforms the same rows. Equal-width quantization likewise fits and transforms the same matrix. The source correctly labels these paths descriptive (`crates/pid-sim/src/offline_harness.rs`, lines 1291–1328 and 4302–4314). Leave-one-out component selection does not make the final same-row projection held out.

The following construction shows why a split is essential. Let $X \in \mathbb{R}^{n\times p}$ be a continuous noise matrix independent of binary target T, with $p \geq n$. With probability one, X has row rank n. Consequently, a target-adaptive linear fit can choose w such that $Xw = T$ on the evaluation rows. Let $\widehat{P}_n$ be the discrete empirical law assigning mass $1/n$ to each fitted pair (Z_i, T_i), where $Z = Xw$. Then

$$I_{\widehat{P}_n}(Z;T) = H_{\widehat{P}_n}(T) \quad \text{on the fitted rows, while} \quad I(X;T) = 0$$

under the population law. This finite-table identity is not a claim that KSG is valid on the resulting tied, singular sample. It does not assert that every PLS call interpolates. It proves that same-row target-adaptive fitting alone cannot support inference.

7.4 Holdout and workable regime

`protocols/holdout_registry_v1.json` records `holdout_count: 0`. The H3 power path also reports missing local PID/CI features, leakage tests, censoring rules, and mandatory baselines (`crates/pid-sim/src/power.rs`, line 1383).

A realistic Prisoma route requires a fixed target, low-cardinality categorical variables or a training-fitted frozen quantizer, disjoint evaluation episodes, a valid row-dependence contract, preregistered endpoints, retained atom signs and abstentions, and one registered confirmatory holdout. High-dimensional learned embeddings, same-row PLS, adaptive bins, and unrestricted continuous PID are outside that regime.

8 Galadriel

8.1 Implemented rule

Galadriel pins pid-core to this revision:

```
1cd2424f7967e1752dcc8e53859e8fdad3566f51
```

The dependency enables `experimental-pipelines`. The pin does not match the pid-rs revision reviewed here.

The PID path is optional and off by default. Signed-correlation analysis remains the default. Galadriel's architecture rules state that `Nominal` cannot create permission and `InsufficientEvidence` cannot become `Nominal` (`AGENTS.md`, lines 117–146). These are correct permanent boundaries.

The decision rule in `crates/galadriel-pid/src/engine.rs`, lines 1283–1527, estimates every gated pairwise MI, builds a threshold from the MI floor and strongest pair, finds the largest MI clique, requires one unique strict majority, and attributes an excluded channel only when all its clique edges are assessed and low.

The I^{sx} redundancy and synergy values are computed separately and never drive the verdict. The accurate name is *pairwise-MI consensus with report-only PID atoms*; it is not a PID verdict.

The code asserts a regular full-dimensional support model and adds deterministic Gaussian pseudo-noise through pid-rs `Jitter`. Current pid-rs states that this primitive changes the law, drops application provenance, and is not generic tie repair. A recorded scale and seed do not prove the population noise model, finite MI, IID rows, or calibration.

8.2 Counterexample: pure synergy

Let $X, Y \sim \text{Bernoulli}(1/2)$ be independent and set $T = X \oplus Y$. Then

$$I(X;Y) = I(X;T) = I(Y;T) = 0, \qquad I((X,Y);T) = \log 2. \tag{3}$$

Conditional on either input, T remains fair. Given both inputs, T is deterministic. The pairwise graph has no positive edge despite exact three-variable dependence. A positive MI floor therefore yields insufficient evidence, and report-only atoms cannot change the verdict.

Equation (3) is a structural counterexample to the decision statistic. Binary data are not offered to the continuous KSG estimator. The result shows that pairwise consensus is not a general synergy detector.

8.3 Counterexample: coherent-majority inversion

Let A, U, E_1, E_2, E_3 be independent standard Gaussian variables and define

$$M_j = A + \sigma E_j \quad (j = 1, 2, 3), \qquad H = U, \qquad \sigma > 0. \tag{4}$$

Every pair (M_i, M_j) is regular jointly Gaussian with $\rho = 1/(1 + \sigma^2)$. Therefore

$$I(M_i; M_j) = -\frac{1}{2}\log(1 - \rho^2) > 0, \qquad I(H; M_j) = 0. \tag{5}$$

For a threshold between these values, the unique strict-majority clique is $\{M_1, M_2, M_3\}$. The rule attributes H as decoupled. If the three coherent channels share one fault or spoof while H is correct, attribution is reversed. The rule identifies agreement, not truth. Geometry and delete-block checks do not provide a trusted reference label.

8.4 Counterexample: targets rotate across atom rows

The atom helper sorts all channels other than row i, uses the first as source two, and uses the mean of the remainder as target. For three modality keys $X < Y < Z$, Table 4 follows directly from the code.

Table 4: Galadriel's three-channel report rows decompose different targets.

Report row	Source 1	Source 2	Target
X	X	Y	Z
Y	Y	X	Z
Z	Z	X	Y

Input list permutation is stable; target rotation is a different semantic question. Let X, Y be independent fair bits and encode $Z = (X, Y)$ as one four-state variable. Then

$$I((X,Y);Z) = 2\log 2, \qquad I((Z,X);Y) = \log 2. \tag{6}$$

Since PID atoms reconstruct the applicable total mutual information, atom vectors cannot remain invariant when the right-hand side changes. Cross-row values are different decompositions and must not be treated as coordinates of one fixed PID.

8.5 Workable Galadriel regime

The current engine can serve as an advisory dependence screen only when scalar common-frame projections and a window are fixed, population support is defensible, row dependence and drift are declared, data pass the estimator gates, thresholds are calibrated for the exact repeated-window policy, and coherent-majority cases are included. It remains non-causal and non-authorizing. It is not a pure-synergy detector, trusted-majority oracle, calibrated attack posterior, or general fault locator.

9 Haldir

9.1 Implementation and stale statements

No direct pid-rs or pid-core dependency was found in Haldir manifests or Rust source. The primary PID artifact is the design and adversarial review `docs/galadriels-mirror.md`; it is not executable integration.

Three statements require correction:

1. Lines 356 and 423 say pid-core has no multiple-comparison correction. Current pid-rs exports the procedures of Benjamini and Hochberg and of Benjamini and Yekutieli under `experimental-pipelines`. Both still require valid input p-values and their stated dependence conditions.
2. Line 270 recommends seeded jitter as a tie/radius repair. Current pid-rs rejects generic jitter repair because added noise changes the estimand.
3. The note repeatedly calls raw block-subsample ranges confidence intervals. Current pid-rs calls them sensitivity diagnostics, not generally calibrated confidence intervals.

The note correctly retains important negative results: slow-drift blind spots, common-majority inversion, the sample-size/latency conflict, and the fact that $I_{\min}$ does not validate I^{sx}. These findings should survive the API update.

9.2 Permanent authority non-expansion

The Haldir authority model says advisory evidence cannot grant or widen authority and a failed or missing conjunct has no fallback. Let $A_0(s,a)$ be Haldir's authority predicate before PID evidence, and let $A_{\mathrm{PID}}(s,a,e)$ read PID evidence e. The permanent invariant is

$$\forall s,a,e: \qquad A_{\mathrm{PID}}(s,a,e) \Longrightarrow A_0(s,a). \tag{7}$$

The post-PID authorized action set must be a subset of the original set. A predeclared restriction $A_0 \wedge R(e)$ is structurally allowed. Any $A_0 \vee G(e)$, override of a failed conjunct, or conversion from an atom to a plant command is forbidden.

Equation (7) remains in force even after formal verification or statistical calibration. Correctness of a statistic does not confer authority. Unavailable PID remains unavailable; it does not become zero, nominal, or permission.

9.3 Workable Haldir regime

PID may enter Haldir only as replayable advisory evidence or as a predeclared restriction that cannot create an allow. Missing, failed, abstained, and numerically unresolved results must remain distinct states. Runtime command selection, effector routing, and safe-action selection remain outside pid-rs.

10 Crebain

10.1 Implemented producer

The current-head refresh from `448c9c43c15b0081c879037aa62b6a9f1bc4a341` to `f1f4c050a1aa6f703b1b25699c232f4305bfb001` changed dependency, security, baseline, and CI metadata but no PID/SxPID, Galadriel scientific contract, sensor-fusion, or authority-path source. The full source/component validation passed in a detached clean worktree. This strengthens the currency of the source reading; it does not change the integration determination.

Crebain has no direct pid-rs dependency. It implements an optional Galadriel evidence producer. Publication requires the off-by-default `ncp` feature and `CREBAIN_GALADRIEL_ENABLE=1`. A successful local put does not prove receiver delivery, decoding, acceptance, or action.

The strongest comparable field is `ConsistencyProjection`. It binds a common frame, context, and frozen-prior identifier. The base observation binds track, time, sequence, modality, NIS, optional innovation and covariance, and the optional projection. This is useful producer provenance; it is not a complete PID adapter.

The envelope does not bind a fixed target and ordered sources, row window and horizon, sampling/dependence model, population support, transform split, PID measure and method identity, estimator settings, signed atom semantics, terminal status, or calibrated uncertainty/multiplicity policy.

Crebain does not execute registry transform chains. A common projection is emitted only when the input frame already matches and the chain is empty. Registry transform, calibration, and projection references are not loaded or verified.

10.2 Missing receiver and next interface

Crebain's producer document itself requires a live Galadriel receiver and cross-route assembler, registry agreement, sequence-gap and heartbeat handling, and receiver-side fault campaigns. None was found. A JSONL parser or NIS smoke test is not end-to-end PID evidence.

The next interface must preserve the raw envelope, create a receiver/window receipt that accounts for every included and excluded sequence, bind one target and ordered sources, attach the complete contract in Section 3, emit pid-rs run-log and software-identity receipts, and test loss, reorder, restart, saturation, drift, target rotation, and coherent-majority cases. Producer, observer, Haldir, command, and plant credentials must remain separate. Crebain's current trust-domain table already assigns Galadriel advisory observation only; that boundary is correct.

10.3 Workable Crebain regime

The current workable claim is producer qualification, not PID qualification. A controlled campaign may show bounded publication, transport, receiver acceptance, sequence-gap accounting, and common frame/context preservation. A PID claim begins only after a separate window receipt binds one target, ordered sources, one row law, one support and transform contract, and one exact pid-rs route. An offline categorical or independently frozen quantized analysis may then be feasible. Continuous ISX still needs its own support, geometry, sampling, and calibration evidence. Until that adapter and evidence exist, Crebain provides candidate inputs only.

11 Permanent negative corpus

Table 5 defines mandatory negative cases. Qualification must check the mathematical result and the terminal status.

Table 5: Cross-ecosystem negative fixtures.

Fixture	Required observation	Required behavior
Fair XOR or parity	Pairwise MI is zero and joint information is positive.	Do not describe the Galadriel verdict as synergy-capable. Report a separate categorical PID result only on a matching route.
Coherent Gaussian majority	Majority edges are positive and honest-to-majority edges are zero.	Report agreement, not honest fault or attack source. Make no authority change.
Target rotation with $Z = (X, Y)$	Total MI changes from $2 \log 2$ to $\log 2$.	Reject cross-row atom comparison as one fixed PID.
Same-row adaptive projection	Fitted-row association can be maximal when test-law MI is zero.	Mark exploratory; held-out qualification fails.
SAFE JSON without support	Python accepts structure; continuous Rust preflight abstains.	Preserve `support_contract_unspecified`; never coerce to zero.
Ties plus generic jitter	A numerical route may run after noise is added.	Report a changed/noised estimand or abstain; never claim repair of the original law.
Common-mode fault	Several wrong channels remain mutually coherent.	Do not infer truth from the largest clique.
Slow drift below window resolution	The screen may remain inside its nominal band.	Record non-detection; do not infer absence of fault or attack.
Missing or failed PID	No scalar exists.	Retain missing/failure status; no nominal default and no authority expansion.
Invalid surrogate p-values	Restricted shifts lack exact randomization validity.	Do not feed the scores to BH/BY as exact p-values.

12 Qualification sequence

Do not collapse heterogeneous evidence into one “confidence” score. Complete and report these layers separately:

1. Cross-language schema conformance with golden and rejection messages.
2. Semantic conformance of target, source, event, lattice, and atom mappings against an exact small categorical oracle.
3. Executable refinement or bounded exhaustive checking of histogram, event, lattice, and atom correspondence.
4. Directed-rounding or ball-arithmetic numerical certification for categorical log terms and signs near zero.
5. Validation of the exact row process, support regime, transform split, drift, and uncertainty procedure.
6. Selection and sequential validation for the full predeclared family and repeated-window policy.

7. Independent hidden challenge after commits and thresholds are frozen, followed by publication of every case.
8. Exact consumer replay with run logs on pinned commits.
9. Machine-checked authority non-expansion and mutations that try to bypass every failed authority conjunct.

The categorical offline path can in principle traverse all nine layers. Continuous two-source PID first needs estimator-specific consistency and calibration evidence in the consumer regime. Full mixed-dimensional continuous PID and authority use remain outside the release path.

13 Final decision

pid-rs addresses part of the ecosystem need. It supplies a strong categorical implementation, restricted continuous research estimators, typed failures, provenance, resource controls, and several validation layers. It does not validate a consumer's target, rows, support, preprocessing, selection, repeated alerts, or authority policy.

The immediate priorities are:

1. repair the Prisoma SAFE schema and add a held-out frozen-transform path;
2. name Galadriel's pairwise rule accurately and retain the XOR, target-rotation, and coherent-majority counterexamples;
3. correct stale Haldir API claims and encode permanent non-expansion;
4. implement Crebain receiver/window/scientific-contract receipts before a PID claim; and
5. qualify pinned consumer routes, not pid-rs in isolation.

Until then, all four routes remain advisory, descriptive, experimental, or blocked as stated. No PID result may authorize an action.

A Evidence paths

Paths are relative to each named repository root. Line numbers refer to the source bases in Table 1.

A.1 pid-rs

- `AGENTS.md`
- `README.md`, lines 550–574
- `KNOWN_LIMITATIONS.md`, lines 159–186 and 470–515
- `crates/pid-core/src/bootstrap.rs`, lines 1–35 and 299–345
- `crates/pid-core/src/pipeline.rs`, lines 1–68, 3703, and 3732
- `crates/pid-core/src/preprocess.rs`, lines 1376–1386
- `ecosystem-capabilities.json` and `ECOSYSTEM_CAPABILITIES.md`
- `method-catalog.json` entries:
  - `multiple-testing.bh-by`
  - `resampling.moving-block-bootstrap`
  - `unsupported.generic-knn-bootstrap-ci`

A.2 Prisoma

- `.gitmodules`

- `AGENTS.md`, lines 8–28 and 107–112
- `protocols/holdout_registry_v1.json`, line 11
- `crates/pid-sim/src/offline_harness.rs`, lines 129–182, 369–384, 1291–1328, 3937–3998, and 4276–4314
- `crates/pid-sim/src/power.rs`, line 1383
- `experiments/safe_adapter/contract.py`, lines 1–14 and 147–211
- `experiments/safe_adapter/convert.py`, lines 376–395
- `experiments/safe_adapter/verify.py`, lines 50–135

A.3 Galadriel

- `AGENTS.md`, lines 117–173
- `Cargo.toml`, lines 63–70
- `crates/galadriel-pid/src/lib.rs`, lines 7–60
- `crates/galadriel-pid/src/engine.rs`, lines 1272–1527, 1582–1640, 1820–1905, and 3710–3745

A.4 Haldir

- `docs/galadriels-mirror.md`, lines 204–219, 255–274, 348–367, and 420–434
- `release/0.9.0/authority-model.json`, lines 26–36 and 47–78

A.5 Crebain

- `AGENTS.md`, lines 84–100 and 128–162
- `src-tauri/src/pid_observation.rs`, lines 45–76 and 180–212
- `docs/GALADRIEL_PRODUCER.md`, lines 1–24, 120–169, and 239–267
- `docs/SYSTEM_CONTEXT.md`, lines 13–20 and 115–138
- `SECURITY.md`, lines 100–103

B Required adapter record

An implementation may encode this record as typed Rust, canonical JSON, or an equivalent format. The semantic fields are mandatory:

Group	Mandatory contents
Source identity	Repository, commit, tree/dirty state, executable/build identity, exact pid-rs identity, method-catalog identity and digest.
Scientific identity	Measure, target, ordered sources, units, frame, horizon, row unit, estimand.
Sampling	IID/block/color/stationary status, dependence justification, selection, drift, missingness and drop accounting.
Support	Alphabet or continuous/mixed declaration; required support floor and its external justification.
Transform	Training/evaluation row identities, frozen parameters, source order, fit and application digests, target-adaptive status.
Estimator	Configuration, metric, neighbour/gauge settings, seed, resources, negative handling, observation-noise declaration.
Outcome	Produced/warning/abstained/failed, signed atoms, interval class, multiplicity family, every retained failure.
Replay and policy	Dataset/config/output/run-log/software hashes, replay result, advisory/no-authority marker, authority non-expansion revision.